

Zhuoyan Yu

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EDUCATION

New York University Shanghai

B.S. in Computer Science; GPA: 3.94

Relevant coursework: Machine Learning, Natural Language Processing, Computer Vision.

Expected May 2027

Shanghai, China

PUBLICATION

CodeFORGE: Framework for Orchestrated Rare Programming Language Generation and Evaluation

Zhuoyan Yu, Minghao Shao, Yijun Shen, Yichen Zhao, Boyuan Chen, Yik-Cheung Tam, Muhammad Shafique.
International Joint Conference on Neural Networks (IJCNN), 2026. [Paper] [Code]

RESEARCH EXPERIENCE

LLM-Generated Test Evaluation

Research Project, Serendipity Research Group

Jun 2026 – Present

NYU Shanghai

- Built a three-model evaluation pipeline comparing same-model and cross-model tests on fixed code candidates, with and without candidate visibility.
- Completed a study over 60 frozen LiveCodeBench problems, retaining 64 partially correct programs across 20 problems; implemented reference-consensus oracles, isolated execution, and reproducible analysis.
- Analyzed fault exposure separately from structural coverage, using paired comparisons, problem-level bootstrap intervals, and missingness and source-balance diagnostics. The primary FE@5 analysis covered 22 programs across 11 problems and did not establish a reliable self-leniency effect.

Ant Technologies U.S., Medical and Healthcare AI Lab

Algorithm Engineer Intern (Research)

May – Aug 2026

- Developed nnU-Net v2 pipelines for cerebral microbleed and lacune detection in multimodal MRI, including spatial alignment, data validation, and custom trainers.
- Improved microbleed lesion-level F1 from 0.533 to 0.557 on 72 VALDO cases using internal five-fold out-of-fold evaluation with cross-fitted thresholds.
- Conducted controlled ablations of voxel spacing, patch/batch size, loss, and foreground sampling; implemented lesion-level matching and free-response ROC analysis.
- Extended the pipeline to 40 lacune cases with dual-rater label analysis and paired comparisons on completed folds; documented training and evaluation status for handoff.

CodeFORGE: Rare and Legacy Code Generation

Research Assistant, Serendipity Research Fellowship

Sep 2025 – Apr 2026

NYU Shanghai

- Built a benchmark of 1,196 verified tasks across 12 rare and legacy languages, using a two-model pipeline that separates canonical solution synthesis from input/output test generation.
- Implemented Docker-based pass@k verification and evaluation with model cascading, feedback-driven resampling, and API cost controls.
- Organized verified tasks as JSONL records with instructions, executable solutions, and tests; implemented post hoc difficulty bucketing for benchmark analysis.

RESEARCH EXPERIENCE (CONTINUED)

LoVUR: Long Video Understanding and Reasoning

Sep 2025 – Jun 2026

Research Assistant, Serendipity Research Fellowship

NYU Shanghai

- Annotated 40+ human assembly/disassembly videos and standardized object, part, and action labels to construct POSET/DAG task graphs and support multimodal LLM evaluation of assembly order.

Agentic Figure Transcription for Scientific Papers

May – Aug 2025

Research Assistant, CSDSE Department

NYU Shanghai

- Developed an academic-figure-to-editable-PPTX pipeline combining OCR, SAM segmentation, region-level VLM classification, and deterministic post-processing.
- Implemented Python microservices for text detection and inpainting, mask generation, region cropping and merging, VLM labeling, and recursive crop-aware segmentation of small objects.

Segment-then-Report: Chest X-Ray Interpretation

May – Aug 2025

Undergraduate Researcher, DURF Program

NYU Shanghai

- Designed a controlled comparison of raw X-ray-to-report and mask-overlay report generation pipelines to study the effect of image-region selection on clinical report quality.
- Implemented lung/chest mask generation with nnU-Net v2 and a torchvision fallback; evaluated reports with BLEU and CheXbert-F1, observed degraded performance with mask overlays, and analyzed possible causes.

PROJECTS

MYGO: Geometry-Aware Depth ControlNet

Sep – Dec 2025

Computer Vision Course Project

- Implemented 18-channel depth conditioning for SD1.5 ControlNet, combining raw depth, 16 ordinal layers, and occlusion boundaries.
- Trained and compared baseline and MYGO models on 50k COCO images for 20k steps, evaluating image quality and depth/occlusion consistency.

Curistro: Learning-by-Teaching

Dec 2024 – Apr 2025

NYU Shanghai DIC Innovation Prize '24

- Engineered an MVP learning-by-teaching web app with an OpenAI-compatible LLM backend, prompt-based quiz generation, structured parsing, and an in-memory datastore.
- Enabled students to teach a topic to an AI and receive quizzes across four dimensions, with instant feedback, answer rationales, and knowledge-map visualization.

ADDITIONAL EXPERIENCE

New York University Shanghai IT Services

Sep 2023 – May 2025

Student Worker

Shanghai, China

- Managed equipment loans and IT asset tracking, resolved user inquiries and basic technical issues, and received the 2025 NYU Shanghai Outstanding Student Worker Award.

Institute of Computing Technology, Chinese Academy of Sciences

Jul – Sep 2024

Intern

- Bench-tested implementations of the Maximal Information Coefficient algorithm and authored a technical report on methodology, findings, and applications in data analysis.

SKILLS

Technical: Python, PyTorch, Hugging Face, OpenCV, Docker, Git.

Languages: Chinese, English.